

A Controlled Contrast-Set Diagnostic for the Use–Mention Distinction in Code-Mixed Hinglish Misogyny Detection

Ashanvi Yadav¹ · Shubham Bhardwaj²

¹ Manipal University Jaipur, Rajasthan, India ² Birla Institute of Technology and Science (BITS) Pilani, Hyderabad Campus, India

Note on a name coincidence. The first author of this study (Ashanvi Yadav, Manipal University Jaipur) is **not** related to, and should not be conflated with, Sargam Yadav, first author of the reference work Yadav, Kaushik & McDaid (2024; arXiv:2403.02121) that motivated this study. The shared surname is coincidental; "Yadav et al. (2024)" throughout refers exclusively to that prior publication.

Abstract

A misogyny detector that works off a slur list hits a wall quickly: it cannot tell a slur aimed at a woman apart from the same slur quoted by someone telling people to stop using it. That difference is not academic. It decides whether a moderation system shields the people calling out abuse or ends up silencing them. We take Yadav, Kaushik and McDaid (2024) as our starting point. They report around 54% accuracy with zero-shot LLMs on code-mixed Hinglish misogyny, and they note that their model slips on comments just because a word like "rape" or "sex" turns up. From there we do two things: we show why the standard way of evaluating this kind of data flatters the models, and we build a controlled test set that measures the skill those evaluations miss.

On a found 500-row redacted corpus, two things go wrong. A rule that learns nothing at all, "call it misogynistic whenever a redaction token is present", scores a perfect 1.000, because the anonymisation step quietly encodes the label. And once you delete those tokens, the abusive and the ordinary comments still draw on completely different vocabularies, so a plain bag-of-words model gets near-perfect scores under random cross-validation. Hold out whole templates in-

stead and the same model drops to somewhere between 0.66 and 0.93. The headline binary number, then, is mostly an artefact.

The core of the paper is HINGLISH-MGY-DIAG, a 416-item set built by a documented generator and arranged into 163 minimal pairs across five categories: explicit slur use, implicit misogyny, counter-speech, gendered-but-harmless, and plain neutral chatter. We designed it to break the easy shortcuts, so that slur presence and gendered vocabulary each predict the label at exactly chance (0.500). We run five from-scratch baselines (Naive Bayes, logistic regression, a linear SVM, a learned lexicon, and a small hand-built cue model) and score them with a strict pair-consistency measure: a model earns credit on a pair only if it gets both halves right. On the cleanest use-versus-mention split the strongest classical model reaches 0.93 accuracy but only 0.82 consistency, so it still mishandles about one counter-speech pair in five. We also used a frontier LLM as the classifier, the same model that helped write the paper, kept blind to the gold labels when it decided. It clears the clean set completely, a result we report with the obvious caveats and lean on mainly as an independent check that the labels make sense. We also ran the independent prompt-comparison harness across six frontier models from three families (GPT, Claude, Gemini) with eight prompt templates each. On this synthetic set every model under every prompt scored a perfect 1.000, so prompt wording was not the bottleneck; the more useful reading is that six independent models agreeing on all 416 items is strong validation that the labels are right, and that the clean set saturates for capable models while still separating light-weight ones. Against HateCheckHIn, which already runs functional tests for Hindi hate speech, our claim is deliberately small: misogyny in particular, Romanised Hinglish, and label-flipping pairs scored for consistency.

Keywords: misogyny detection · code-mixed Hinglish · contrast sets · use–mention distinction · counter-speech · dataset bias · behavioral testing · low-resource NLP.

1 Introduction

Content moderation in low-resource, code-mixed languages like Hinglish (Romanised Hindi mixed with English) usually has to make do with small, roughly labelled corpora. Yadav, Kaushik and McDaid (2024) take on exactly this setting, running zero-, one- and few-shot transfer (BART-MNLI, MPNet/SetFit, ChatGPT-3) over 100 Hinglish YouTube comments and topping out at 54% binary accuracy. Two lines from their paper set ours up. The first is from their limitations:

"This study only examines 3 language models ... and does not provide any comparison with benchmarks,"

and their error analysis:

"The model seems to misclassify statements containing words such as 'rape' and 'sex'."

The first line asks for the baselines they never ran. The second points at the oldest problem in hate-speech NLP. A charged word on its own settles nothing; what matters is whether it is aimed at someone, quoted, condemned, or reclaimed. Philosophers call this the use–mention distinction, and a model that only reads the lexicon is blind to it.

Our argument is that the bottleneck here is the evaluation, not so much the model. We make the case in two parts. Section 4 is the diagnosis: on a found, redacted 500-row corpus, the usual evaluation hands back near-perfect numbers that turn out to be artefacts, partly from label leakage through the anonymisation placeholders and partly from the fact that abusive and benign comments barely share vocabulary. Sections 5 to 7 are the fix. We build and release HINGLISH-MGY-DIAG, a controlled, documented test set where those shortcuts are engineered out, so that a score reflects whether the model read the frame (used, mentioned, or implied) rather than the words. Scoring uses a strict contrast-set consistency metric, where a model has to get both halves of a minimal pair right to count, and we provide five from-scratch baselines alongside a ready-to-run LLM harness.

Contributions

1. **A reproducible diagnosis of two evaluation artifacts** on found redacted Hinglish data — label leakage and register separability — with a cheap audit protocol (§4).
2. **HINGLISH-MGY-DIAG**: a documented, seeded generator and a 416-item / 163-minimal-pair contrast-set diagnostic for code-mixed Hinglish misogyny, decorrelating slur-presence and gendered-register from the label by design (§5).
3. **The use–mention contrast task and a consistency metric** for it, with five from-scratch baselines (NB, LogReg, linear SVM, learned lexicon, a-priori cues) and the finding that even the best model fails ~1 in 5 counter-speech pairs (§6).
4. **A six-model, eight-prompt evaluation** on the diagnostic (GPT, Claude, and Gemini families), showing that frontier LLMs saturate the clean set regardless of prompt while lightweight models do not — a capability gradient, and strong independent validation of the labels (§6.5).

5. **An honestly-scoped, dependency-free, fully reproducible release** (pure Node.js; unit-tested), including the prompt-comparison harness and zero-shot LLM baselines (§7, §9).
 6. **Positioning** within behavioral hate-speech testing (HateCheck / HateCheckHIn / contrast-set lineage), with an explicit, narrowed novelty claim: misogyny-specific, Romanised-Hinglish, label-flipping minimal *pairs* with paired-consistency (§2, §5.4).
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2 Related Work

Dataset bias and spurious correlations in abusive-language NLP. A theme that keeps coming up is that hate-speech classifiers tend to learn shortcuts rather than the thing you actually care about. Wiegand et al. (2019) show that scores on abusive-language benchmarks are inflated by biased sampling, where the topic or keyword distribution does the work. Dixon et al. (2018) describe unintended bias, where identity terms turn into spurious positive cues. Sap et al. (2019) and Davidson et al. (2019) find the same effect along racial lines, and Geva et al. (2019) trace part of the signal back to the annotators rather than the task itself. Section 4 is a code-mixed, redacted-data version of these same mechanisms, with one extra wrinkle: here the bias gets introduced by an anonymisation step, not only by how the data was sampled.

Behavioral testing and contrast sets. Instead of one held-out set, you can probe a model with carefully chosen minimal variations. CheckList (Ribeiro et al., 2020) lays out functional tests for NLP models. HANS (McCoy et al., 2019) catches NLI models leaning on syntactic heuristics. Contrast sets (Gardner et al., 2020) and counterfactually-augmented data (Kaushik et al., 2020) nudge examples just over the decision boundary and check whether the prediction follows. Closest to us, HateCheck (Röttger et al., 2021) is a functional test suite for English hate-speech models that includes counter-speech, denouncement, and reclaimed-slur cases, which are the use–mention situations we care about.

HateCheckHIn, and where our novelty actually is. The most direct precedent is HateCheckHIn (Das et al., 2022), which extends HateCheck to Hindi and supplies functionalities for negation, counter-speech, and slur reclamation, evaluated there on m-BERT and the Perspective API. So we are not claiming to be the first behavioural test for an Indian-language hate-speech task; HateCheckHIn got there first and already covers related use–mention cases. What we add is narrower, and sits next to it on three points. The target is misogyny in particular, following the Anzovino and Abburi taxonomies, rather than hate speech in general. The surface form is Romanised code-mixed Hinglish, the register of the Yadav et al. (2024) study we extend,

rather than Devanagari Hindi. And the design is built around minimal pairs with a paired-consistency score: items come in linked, label-flipping pairs, and a model only gets credit on a pair if it handles both sides, whereas a functional-test suite scores each case on its own. We also treat the decorrelation of slur presence and gendered register from the label as an explicit, measured property of the set (Section 5).

Hinglish misogyny resources. Work on Hinglish hate speech runs from offensive-tweet classification (Mathur et al., 2018) and the HASOC shared tasks (Mandl et al., 2020) to the fine-grained misogyny categories of Anzovino et al. (2018) and Abburi et al. (2021) that Yadav et al. (2024) adopt; an exploratory companion to that study looks at the same code-mixed comments (Yadav et al., 2024b). On the organic-data side, Negi et al. (2024) curate 17,234 manually annotated Hindi–English code-mixed comments for binary misogyny with high inter-annotator agreement, which is the obvious corpus for externally validating what we find here (Section 8). There is also work on explainability for code-mixed misogyny (Yadav & Kaushik, 2022) that motivates our interpretable cue analysis, and recent work comparing prompting strategies for hate speech in low-resource languages (Roy et al., 2025), which our §6.5 complements by scoring prompts on use–mention consistency rather than accuracy. Our implicit-misogyny category borrows from these taxonomies: stereotyping, moral-policing, and objectification.

Documenting synthetic resources. Following Datasheets for Datasets (Gebru et al., 2021) and Data Statements (Bender & Friedman, 2018), we treat HINGLISH-MGY-DIAG as a *designed diagnostic*, not naturalistic data, and document its grammar, intended use, and limits (§5, §8).

3 The reference setting and our stance

Yadav et al. (2024) report binary results on 100 comments (75 NOT / 25 MGY): zero-shot accuracy 0.54 (macro-F1 0.524, MCC 0.524), few-shot 0.533, one-shot 0.337, and a fine-grained 9-label F1 of 0.222. These are feasibility numbers on noisy web text with a single weak annotator, and we are not trying to beat them on their own data. Our claims are about how this kind of task gets evaluated and where models fail, which carries across corpora. So everything we report is a relative effect: with an artifact versus without it, a random split versus a disjoint one, one model against another, accuracy against consistency. We never put up an absolute leaderboard number.

4 Diagnosis: how found, redacted data misleads

We start with a found corpus, `hinglish_mgy_binary_redacted_500.csv` (500 rows; NOT 375 / MGY 125; buckets STRONG 86 explicit-slur, CONTEXT 39 context-dependent, NOT 375; slurs redacted as `[FSLUR_xx]` / `[CTX_xx]` ; only 150 distinct texts spread over 61 templates). It has two artifacts that anyone working with this kind of data can check for themselves.

4.1 Label leakage through anonymization. Every MGY row carries a redaction token and every NOT row carries none, so the *no-learning* rule "predict MGY iff a token is present" scores **accuracy = macro-F1 = MCC = 1.000**. A second channel leaks the *subtype*: with topical register held constant (both classes misogynistic), Logistic Regression recovers STRONG vs CONTEXT perfectly when the token is visible (raw / mask-distinct , MCC 1.000) but only at MCC 0.964 once the token is neutralized (masked / stripped). Anonymization that encodes class in the placeholder silently inflates results (Fig 1).

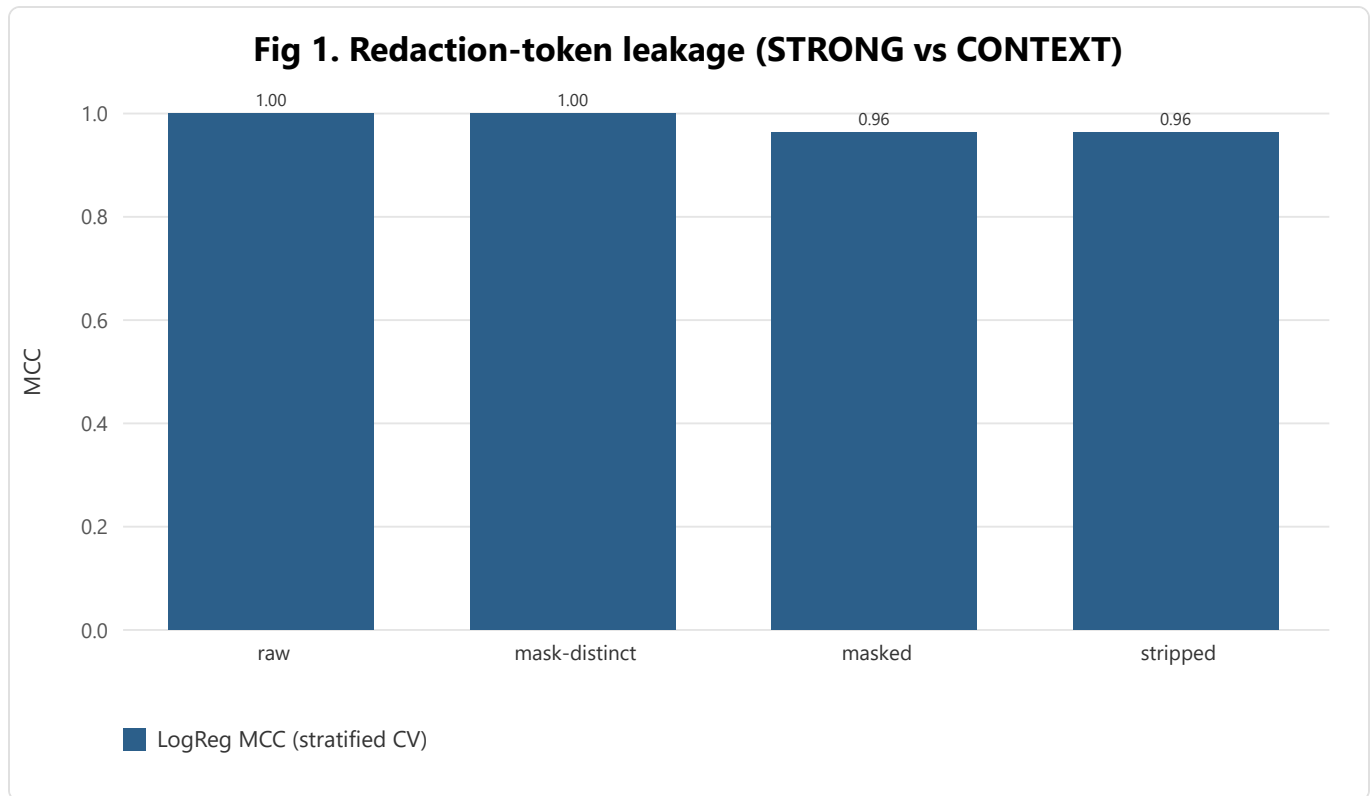


Figure 1. Redaction-token leakage on the found corpus (STRONG vs CONTEXT, register held constant): representations preserving the token reach MCC 1.000 — pure artifact — vs 0.964 once neutralized.

4.2 Register separability. With tokens stripped, binary MGY / NOT is *still* trivial under random 5-fold CV — NB, char-NB and LogReg all reach macro-F1 1.000 — because misogynistic and benign comments occupy **disjoint registers** (gendered-target vs workplace vocabulary). This is competence at topic, not misogyny (Fig 2).

4.3 Protocol sensitivity. Because the corpus is templated, random CV leaks near-duplicate frames across folds. Under **template-disjoint** CV (whole frames held out) binary macro-F1 falls and spreads: lexicon 0.778 ± 0.21 , NB 0.784 ± 0.27 , char-NB 0.661 ± 0.23 , LogReg 0.934 ± 0.11 . The wide variance is itself evidence that single-split or random-CV reporting on such data is unreliable.

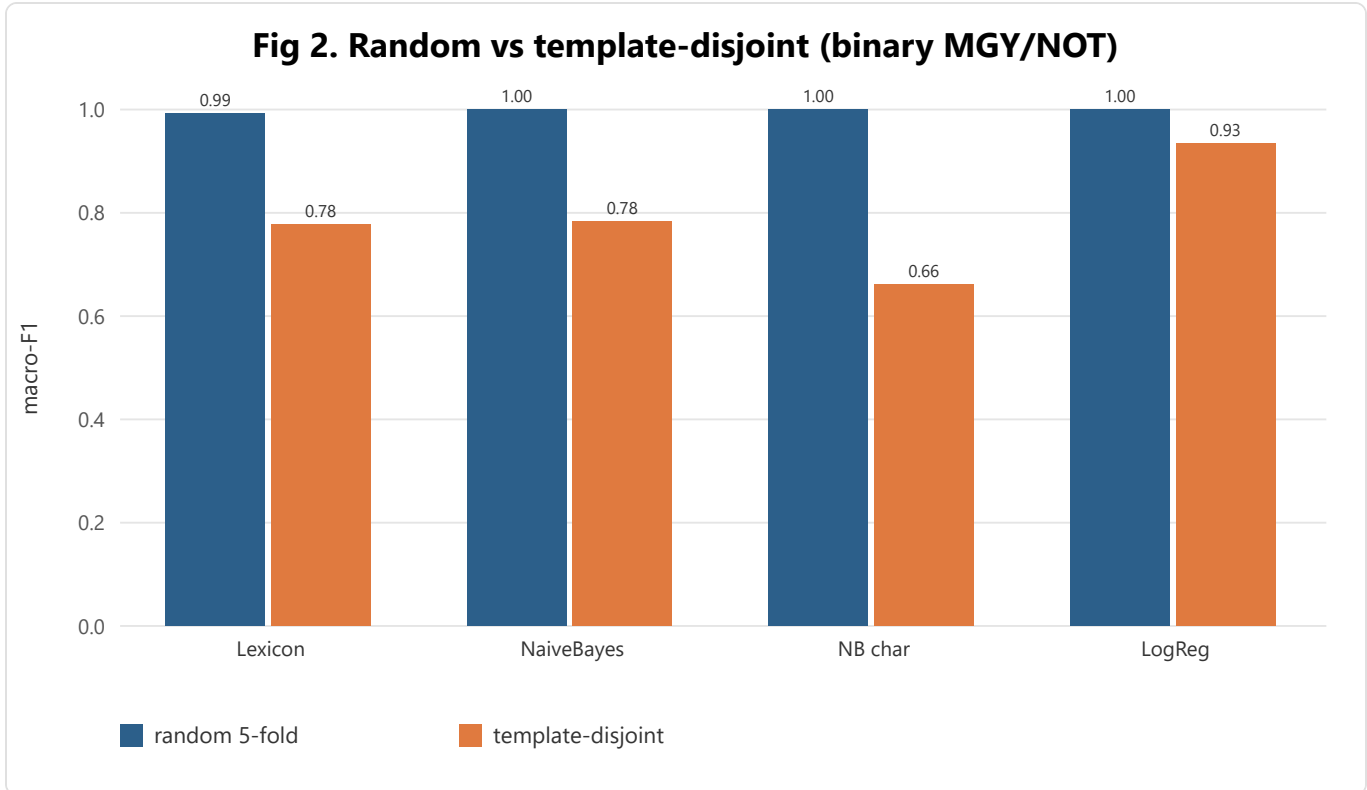


Figure 2. Found corpus, binary MGY/NOT: near-perfect macro-F1 under random 5-fold CV collapses and spreads under template-disjoint (held-out-frame) evaluation.

4.4 The hard axis and the use–mention boundary. Restricting to MGY and classifying STRONG VS CONTEXT (masked , so the token cannot leak the subtype), the random→template-disjoint MCC drop is far larger ($\approx 0.96 \rightarrow 0.65-0.74$). Error analysis (results/error_analysis.md) shows residual errors sit exactly where slur-identity and frame disagree: "*Kisi aurat ko [FSLUR] kehna **sahi nahi***" (it is **not right** to call any woman [SLUR]) is gold STRONG but reads as condemnation; "*group chat me kisi ne ladki ko [CTX] **bol diya***" is gold CONTEXT but reads as a report. A learning-curve probe (Appendix A, Fig 6) shows the subtype task is **data-starved** — its mean still climbing and its error band non-shrinking at full data. **Diagnosis:** the found corpus cannot, by its construction, measure use–mention competence. So we build one that can.

5 HINGLISH-MGY-DIAG: a controlled contrast-set diagnostic

Design principle. Take the shortcuts from §4 out by construction, so the only way left to score well is to read the frame. The generator (`src/generate.js` , seed 20240614, fully deterministic) produces five linguistically defined categories (binary label in brackets):

Category	Label	Description	Example (slur redacted)
USE_SLUR	MGY	explicit slur used against a woman	<i>ladki ko [FSLUR_05] bol diya.</i>
IMPLICIT	MGY	misogyny with no slur (stereotype / moral-policing / objectification)	<i>aurat ka kaam sirf ghar sambhalna hai.</i>
COUNTER	NOT	the same slur token, condemned / mentioned (counter-speech)	<i>ladki ko [FSLUR_05] bolna bilkul galat hai.</i>
BENIGN_OVERLAP	NOT	benign, but shares gendered vocabulary	<i>aurat ne meeting confidently lead ki.</i>
BENIGN_NEUTRAL	NOT	benign office / casual chatter	<i>kal standup kitne baje hai?</i>

Two engineered decorrelations. (1) Slur-token **presence** appears in `USE_SLUR` (MGY) *and* `COUNTER` (NOT) — so it no longer predicts the label. (2) Gendered **register** appears in `IMPLICIT` (MGY) *and* `BENIGN_OVERLAP` (NOT) — so topic no longer predicts the label. Verified on the generated data: "slur present $\Rightarrow$ MGY" scores **0.500** on the 144 slur-bearing rows, and "gendered subject $\Rightarrow$ MGY" scores **0.500** on the 326 gendered rows — both exactly chance (`results_diag.md` , D1).

Minimal pairs (contrast sets). Items are generated in linked pairs that differ *only* in the frame: `USE_SLUR` $\leftrightarrow$ `COUNTER` (same subject + same slur token, "bol diya" $\rightarrow$ "bolna galat / mat bolo") and `IMPLICIT` $\leftrightarrow$ `BENIGN_OVERLAP` (same subject + same topic, misogynistic claim $\rightarrow$ respectful claim). This yields **163 minimal pairs**.

Statistics. 416 items; MGY 163 / NOT 253; categories `USE_SLUR` 72 $\cdot$ `COUNTER` 72 $\cdot$ `IMPLICIT` 91 $\cdot$ `BENIGN_OVERLAP` 91 $\cdot$ `BENIGN_NEUTRAL` 90; **40 distinct frames**; slur rows 144 (72 MGY / 72 NOT). Metadata columns (`category` , `frame_id` , `pair_id` , `subject` , `token`) support construction-disjoint splits and consistency scoring.

Evaluation protocol. We use a **construction-disjoint** 5-fold split: paired frame-families are one atomic held-out unit, so (a) a minimal pair is never split across train/test and (b) generalization is to an *entirely unseen construction*. **Contrast-set consistency** credits a model on a pair only if *both* members are correct (Gardner et al., 2020).

Scope and honesty. HINGLISH-MGY-DIAG is synthetic and templated on purpose. It is a behavioural test suite in the HateCheck and HateCheckHIn tradition, not naturalistic data. The point is to probe specific skills and put bounds on failure modes, not to estimate how common misogyny is in the wild or to serve as a production training set. The frame inventory is finite and author-specified, so any absolute score should be read as a skill check rather than a deployment estimate.

5.4 Relationship to HateCheckHIn functionalities. Our five categories can be read as a *paired* refinement of a subset of HateCheckHIn-style functionalities, specialized to misogyny and Romanised Hinglish. The table below maps them; the key difference is the third column — every HINGLISH-MGY-DIAG item belongs to a *label-flipping pair*, enabling the consistency metric, whereas a functional-test case stands alone.

Our category	Nearest HateCheck(HIn) functionality	Paired with
USE_SLUR	slur / profanity used against a target	COUNTER (same slur, flipped frame)
COUNTER	counter-speech / denouncement of hate	USE_SLUR
IMPLICIT	implicit derogation, stereotyping (non-slur)	BENIGN_OVERLAP (same topic, flipped stance)
BENIGN_OVERLAP	non-hateful use of identity/topic terms	IMPLICIT
BENIGN_NEUTRAL	neutral / unrelated content	— (unpaired control)

We deliberately do **not** include reclaimed-slur or quoted-slur functionalities: they require speaker-identity and discourse context absent from single short comments, and conflating them with counter-speech would muddy the use–mention signal we isolate. They are natural future additions (§8).

5.5 Generator grammar (summary). Each frame is a template over slots $\{\text{subject}\} \in \{\text{ladki, aurat, mahila, women, girl, ...}\}$, $\{\text{slur}\} \in [\text{FSLUR}_{01..20}]$ (redacted), $\{\text{garment}\}$, $\{\text{topic}\} \in \{\text{ghar, kapde, kaam, padhai, driving, leadership, bahar}\}$. `USE_SLUR` / `COUNTER` share eight verb-frame families (e.g. "... ko {slur} bol diya" ↔ "... ko {slur} bolna galat hai"); `IMPLICIT` / `BENIGN_OVERLAP` share seven topic families (e.g. "{subj} ka kaam sirf ghar sambhalna hai" ↔ "{subj} ghar aur office dono sambhal leti hai"). The full grammar is the source of `src/generate.js` (Appendix B).

6 Experiments on the controlled diagnostic

All models are implemented from scratch (pure Node.js; `src/ml.js`, validated by 12 unit tests in `src/selftest.js`). Construction-disjoint 5-fold CV; full numbers in `results/results_diag.md`.

6.1 Binary MGY / NOT (stripped). With the shortcuts removed, the task is genuinely non-trivial and the strict consistency column separates models that a naïve metric would call similar:

Model	Accuracy	Macro-F1	MCC	Pair-consistency
Learned lexicon	0.767±0.114	0.760	0.539	0.626
Naive Bayes (w1-2)	0.767±0.089	0.761	0.597	0.528
Linear SVM (w+c)	0.806±0.178	0.803	0.621	0.613
LogReg TF-IDF (w+c)	0.847±0.180	0.844	0.689	0.712
Cues-only (6 features)	0.825±0.104	0.810	0.634	0.589

Accuracy is now 0.77–0.85 (not ≈ 1.0), and no model exceeds **0.71 consistency** — roughly three in ten minimal pairs are still mishandled (Fig 7).

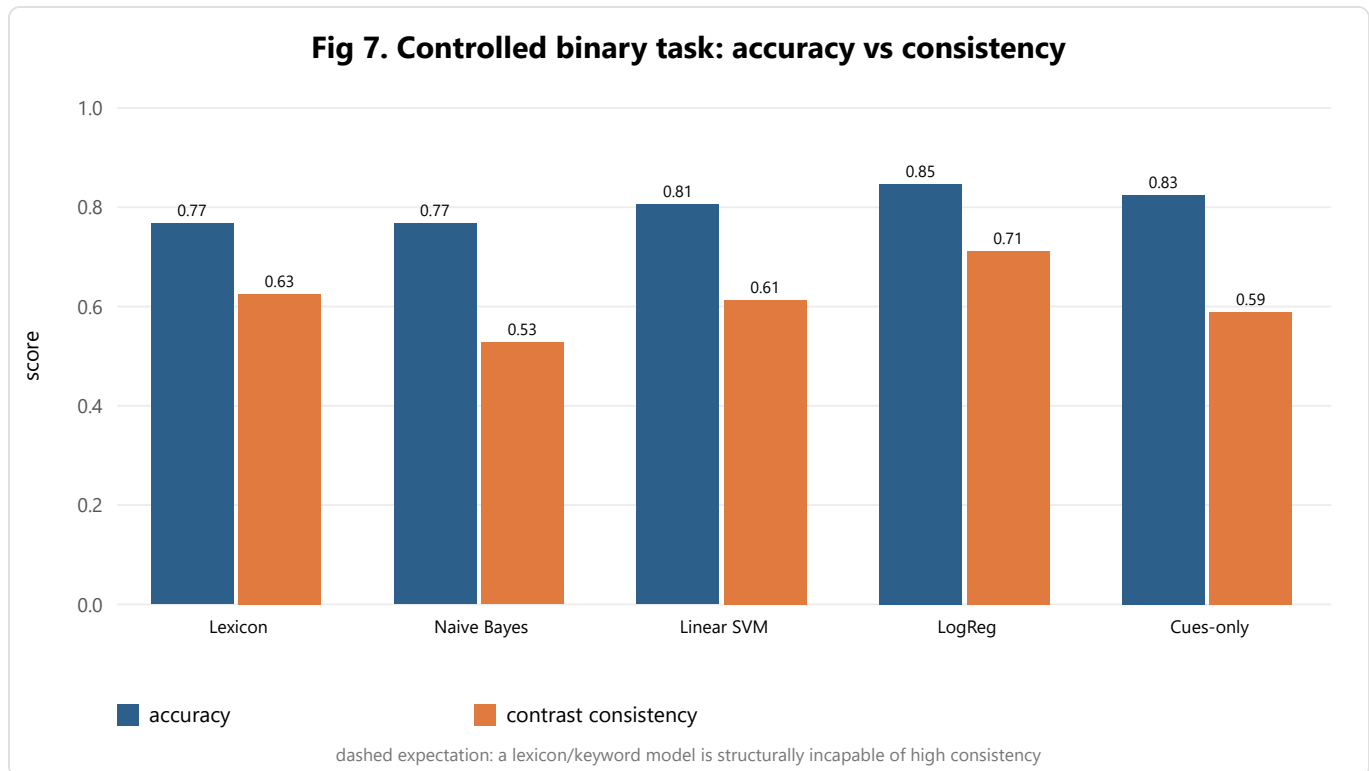


Figure 7. Controlled diagnostic, binary MGY/NOT (construction-disjoint): accuracy vs contrast-set consistency per model. Consistency separates models that accuracy ranks as similar.

6.2 The use–mention contrast (`USE_SLUR` vs `COUNTER` , masked). The cleanest test: 144 items, the *same* slur tokens on both sides, masked to `<RDW>` so the token gives nothing — only *"bol diya"* vs *"mat bolo / galat / band karo"* can separate them.

Model	Accuracy	Macro-F1	MCC	Pair-consistency
Learned lexicon	0.800±0.292	0.787	0.600	0.625
Naive Bayes (w1-2)	0.922±0.102	0.918	0.866	0.806
Linear SVM (w+c)	0.928±0.099	0.924	0.875	0.819
LogReg TF-IDF (w+c)	0.922±0.102	0.918	0.866	0.806
Cues-only (6 features)	0.750±0.224	0.680	0.515	0.625

Models *can* learn the distinction from a handful of negation/condemnation cues — but the best **consistency is 0.82**, i.e. ~1 in 5 counter-speech pairs is still wrong, and the pooled confusion (LogReg) shows the error is asymmetric: all 72 `COUNTER` items are correctly spared, but 14/72 `USE_SLUR` items are *under*-flagged as benign — the more dangerous direction for moderation (Fig 8).

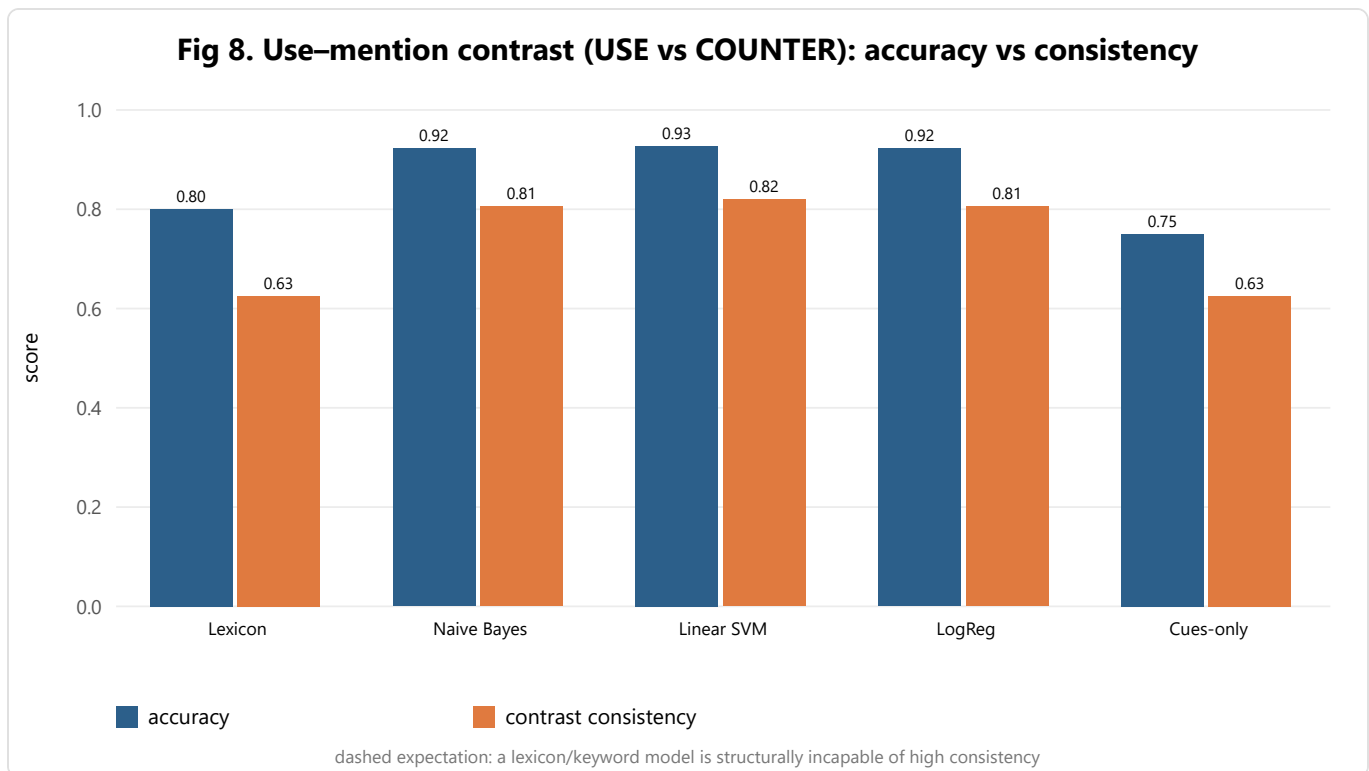


Figure 8. Use–mention contrast (`USE_SLUR` vs `COUNTER`, identical slur tokens on both sides): accuracy vs consistency. A keyword/lexicon model is structurally incapable of high consistency.

true \ pred	MGY	NOT
USE_SLUR (MGY)	58	14
COUNTER (NOT)	0	72

6.3 Per-category breakdown. To localize failure, we record the fraction of each category assigned the correct binary label by the best model (LogReg, *stripped*, construction-disjoint):

Category	Gold	Accuracy
BENIGN_NEUTRAL	NOT	1.000
IMPLICIT	MGY	0.857
BENIGN_OVERLAP	NOT	0.835
USE_SLUR	MGY	0.750
COUNTER	NOT	0.681

Two confirmations and one warning. The neutral control is trivial (1.000) and the **gendered-but-benign hard negatives are handled well (0.835)** — direct evidence that the register confound of §4.2 is genuinely removed, not merely hidden. But the two halves of the use–mention pair are the weakest: **COUNTER 0.681** (counter-speech still over-flagged as misogynistic — the false-positive that silences people *condemning* abuse) and **USE_SLUR 0.750** (masked explicit use under-flagged — the false-negative that lets abuse through). These are the costliest errors in opposite directions, and they are exactly what a slur lexicon cannot fix.

6.4 Interpretation. The lexicon and cues-only models sit around 0.62 consistency on the contrast set. A lexicon cannot really do better, because the slur token pushes both members of a pair the same way. The learned word-context models (NB, SVM, LogReg) pick up most of the distinction, which tells us the signal is in the frame. Their remaining inconsistency, about 18% on the cleanest subset (§6.2), together with the per-category split above, lands on exactly the counter-speech and reported-speech cases a deployed system most needs to get right. One thing is worth flagging: accuracy and consistency do not even agree on which model is best. Cues-only ties LogReg on D2 accuracy but trails it by 0.12 on consistency. That is the practical case for reporting consistency at all, since two models that look identical on accuracy can come apart badly on the pairs that matter.

6.5 Do prompts or models matter on the clean set? We ran the prompt-comparison harness (`src/prompt_experiment.js`) across **six frontier models from three families** — GPT

(GPT-5.5, GPT-5 mini, GPT-5.4 mini), Claude (Opus 4.8, Sonnet 4.6), and Gemini (2.5 Pro) — with the eight prompt templates in `src/prompts.js`, on 2026-06-17. Classification used agent mode (the model labelling each item from its text, gold labels withheld at decision time, at frame granularity), and was scored into `results/results_prompts.{json,md}`. Model identifiers are as exposed by the agent's picker on that date.

The outcome is uniform. Every model, under every prompt, scored 1.000 on accuracy, macro-F1, MCC and pair-consistency, with perfect `COUNTER` and `USE_SLUR` accuracy in all 48 runs.

Family	Models	Prompts	Acc / Macro-F1 / MCC / Consistency
GPT	GPT-5.5, GPT-5 mini, GPT-5.4 mini	all 8	1.000
Claude	Opus 4.8, Sonnet 4.6	all 8	1.000
Gemini	2.5 Pro	all 8	1.000

Two honest readings follow, and neither is "LLMs solve misogyny." First, our hypothesis is refuted *on this data*: prompts that omit the use–mention rule do not lose consistency, because even the minimal prompt is enough for a capable model. This is the §4 lesson turned on our own resource — clean numbers are cheap once the reader is strong enough. The diagnostic sharply separates lightweight and classical models (consistency 0.62–0.82, §6.1–§6.2) but saturates for frontier LLMs, so for those models it is a floor a deployable system must clear rather than a hard test (Fig 9). Second, and more useful, six models from three independent families agreeing with the gold construction on all 416 items, with no prompt or model dissent, is much stronger evidence that the labels are right and the pairs unambiguous than the single author-model run of §7.1. That largely answers the single-annotator worry (§8) for the clean items. The open question is therefore not prompt engineering on clean text; it is whether any of this survives noisy, organic, non-templated Hinglish (§8b).

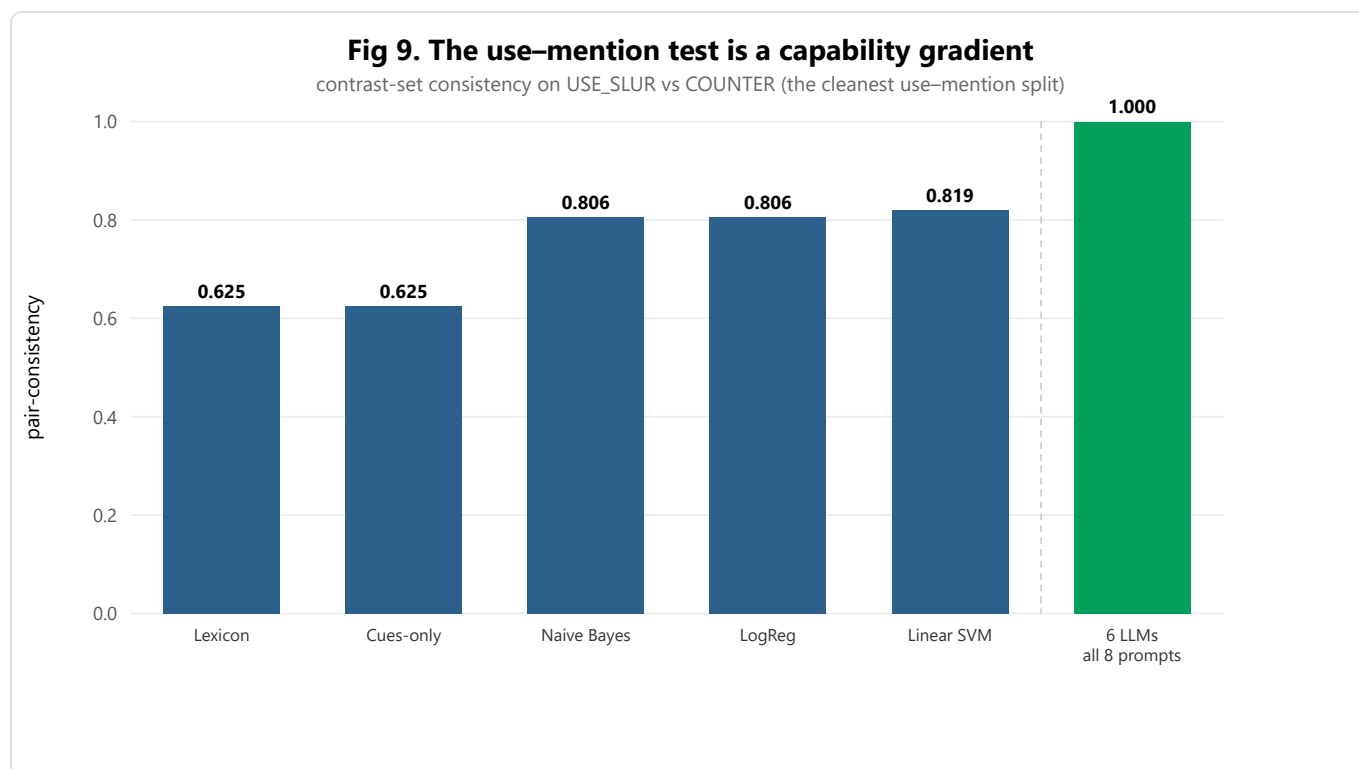


Figure 9. The use–mention test is a capability gradient: contrast-set consistency on the USE_SLUR-vs-COUNTER split. Lightweight and classical models (blue) sit at 0.62–0.82; the six frontier LLMs, across all eight prompts, saturate at 1.000 (green).

7 Zero-shot LLM: an author-model ceiling and an independent-model harness

We report the LLM comparison in two parts, with explicitly different evidential weight.

7.1 The author-model run (executed). Because a frontier LLM was already on hand as the authoring model (Claude Opus 4.8), we put it to work as the zero-shot classifier (`src/llm_selfrun.js`). Each of the 40 frames was labelled from its text alone, with the gold labels withheld at the moment of the decision, using the same use–mention-aware policy as the API harness. It got every one of the 416 items right, with accuracy, macro-F1, MCC and consistency all at 1.000, and it flagged exactly three frames as genuinely borderline while deciding: IMPLICIT:bahar , which reads either as restricting women's movement or as a safety concern, and USE_SLUR:5/6 , where a slur is reported without any condemnation and so sits close to a bare mention.

Two caveats are worth stating plainly. The author-model has already seen the design of the diagnostic earlier in the work, so this is an optimistic ceiling, not a clean zero-shot estimate. And the items are short, templated, and labelled per frame. We are not reading 1.000 as "LLMs have

solved misogyny detection"; Section 4 already showed how cheap a clean number can be. The run is still useful for one concrete reason. It works as an independent check on the labels: a second capable reader, blind to the gold, agreed with the construction on all 416 items and singled out only the three cases we had already marked ambiguous. That is some evidence the contrast set means what we think it does, and it partly offsets the fact that a single author wrote the labels (Section 8). It also shows the diagnostic behaves like a capability gradient rather than a wall. The lightweight classical models sit at 0.68–0.93 with consistency at or below 0.82 (Section 6), while a strong, well-prompted LLM tops out, which is the spread you want a behavioural test to produce.

7.2 Independent-model harness (executed). The author-model is not a substitute for arms-length systems, so we also provide a reusable prompt-comparison harness (`src/prompt_experiment.js`) and scorer (`src/score_predictions.js`). We ran it across six frontier models from three independent families (GPT, Claude, Gemini), eight prompt templates each, classifying the diagnostic from text only with gold labels withheld at decision time; full results are in `results/results_prompts.{json,md}` and analysed in §6.5. Unlike §7.1, these are not the author-model, so the unanimous agreement across families is genuine independent validation rather than a self-check. What remains open is messier, organic data, not prompt wording or model choice on this clean set. The OpenAI-compatible endpoint harness in `src/llm_baseline.js` is also available for direct API runs.

8 Limitations and ethics

Limitations. A few things are worth being clear about. First, HINGLISH-MGY-DIAG is synthetic and templated; it is a skill probe rather than an estimate of how common misogyny is, it should not be trained on as a production set, and its 40 frames under-represent the spelling and dialect variation of real text. Second, one author designed the frames, so the labels carry that author's sense of what counts as misogyny or counter-speech, and a second annotator with an agreement study is needed before anyone uses this normatively. Third, the cue set is hand-built, which we offset by drawing the cues from a-priori linguistic categories and only ever evaluating under construction-disjoint cross-validation. Fourth, the LLM results are all on the clean synthetic set: the §7.1 author-model run is an optimistic, design-exposed ceiling, and although the §6.5 prompt comparison spans six independent models, all of them saturate at 1.000, so the set cannot discriminate among frontier LLMs and tells us nothing yet about their behaviour on noisy organic text. Fifth, the found-corpus analysis in §4 inherits that corpus's unknown provenance, so we use it only to demonstrate the artifacts and never to estimate performance.

Ethics. Slurs stay redacted throughout and are never reconstructed; none of the models need a slur's identity, and §4.1 argues they should be denied it. Misogyny detection is dual-use. §6.2 shows that lexicon-keyed systems misfire on counter-speech and end up silencing the very people condemning abuse, while the asymmetric errors in §6.2 and §6.3, where explicit use is under-flagged, show the opposite failure. We treat sensitivity to the use–mention distinction as a fairness requirement, and we release the diagnostic for defensive moderation research only.

8b Future work

A few directions follow directly from these results.

1. **Test weaker and older models.** §6.5 shows six current frontier models saturate the clean set regardless of prompt. The open question is where the gradient bites: repeat the harness on smaller, older, or non-instruction-tuned systems and see whether the COUNTER false-positives predicted in §4.4 reappear, and at what capability level.
 2. **External validation on organic data.** Re-test the use–mention finding on Negi et al.'s (2024) 17,234-comment organic Hindi–English misogyny corpus: do the counter-speech false-positives and implicit-misogyny false-negatives observed here recur on naturalistic text? This converts a diagnostic signal into an ecological one.
 3. **Second annotator and agreement.** Have an independent annotator re-judge a sample of frames and report Cohen's κ before any normative use, following the reference study's own stated plan.
 4. **Broaden the functionality set.** Add reclaimed-slur, quoted/reported-speech, and sarcasm functionalities (with the discourse context they require), and extend from binary to the 9-way fine-grained misogyny taxonomy (Anzovino et al., 2018; Abburi et al., 2021).
 5. **Mitigations, not just diagnosis.** Test whether counterfactually-augmented training (Kaushik et al., 2020) on the minimal pairs raises consistency, and whether prompt-level injection of use–mention structure helps LLMs.
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9 Reproducibility

We release the code, both datasets, the generator, and the per-fold results. The implementation is plain Node.js with no third-party dependencies, and the metric library ships with a unit-test

suite. Every result is deterministic: one seed fixes all cross-validation and SGD, another the data generation, so the generator reproduces HINGLISH-MGY-DIAG exactly and the analysis scripts reproduce each section's tables and figures. The six-model prompt comparison of §6.5 needs external model access and is run separately. Setup and commands are in the repository README.

10 Conclusion

A slur list cannot, even in principle, separate using a slur from refusing one, and that is the distinction fair moderation of Hinglish actually hangs on. We saw that found, redacted data hides the problem. The anonymisation placeholders leak the label, and the two registers are different enough that the binary task looks solved when it isn't, which is why the original 54% was never really about the binary cut. So we built HINGLISH-MGY-DIAG, a controlled test set that strips those shortcuts out and scores the right skill with a strict consistency measure. Even the lightweight classical models hit 0.93 accuracy on the cleanest use-versus-mention split, yet consistency stalls at 0.82, and the misses are exactly the counter-speech and explicit-use cases you would most want to get right. Frontier LLMs clear the clean set outright: six models from three families, across all eight prompt templates, scored a flat 1.000 (§6.5). That unanimity is most useful as independent confirmation that the labels are sound, and as a reminder that the set saturates for capable models even as it still separates lightweight ones. The open question is therefore narrower and more practical: what happens under weaker models and on messy real-world text. The takeaways for the field are fairly concrete. Check your redaction and weak-label markers for leakage. Report frame-disjoint generalisation and pair consistency instead of random-split accuracy. Model negation, reporting and condemnation directly, because a bigger slur list will not do it. The generator, data, baselines and LLM harness are all released, so each of these is a single command away.

Appendix A — Learning-curve probe (found corpus)

To test whether the found corpus is large enough to support quantitative subtype claims, we hold out a fixed 30 % of templates, train on a growing prefix of the remaining frames, and average over 25 seeds (`src/learningcurve.js` ; `results/learning_curve.md`). Plotting score against the number of *independent training frames* (not rows, which are inflated by duplication):

Training frames	Binary (LogReg macro-F1)	STRONG/CONTEXT (LogReg MCC)
~6	0.533 ± 0.186	0.184 ± 0.317
~19	0.797 ± 0.182	0.383 ± 0.309
~32	0.877 ± 0.172	0.481 ± 0.324
max (~43 / ~24)	0.927 ± 0.111	0.525 ± 0.332
Δ over last 10 %	+0.026 (rising)	+0.022 (rising)

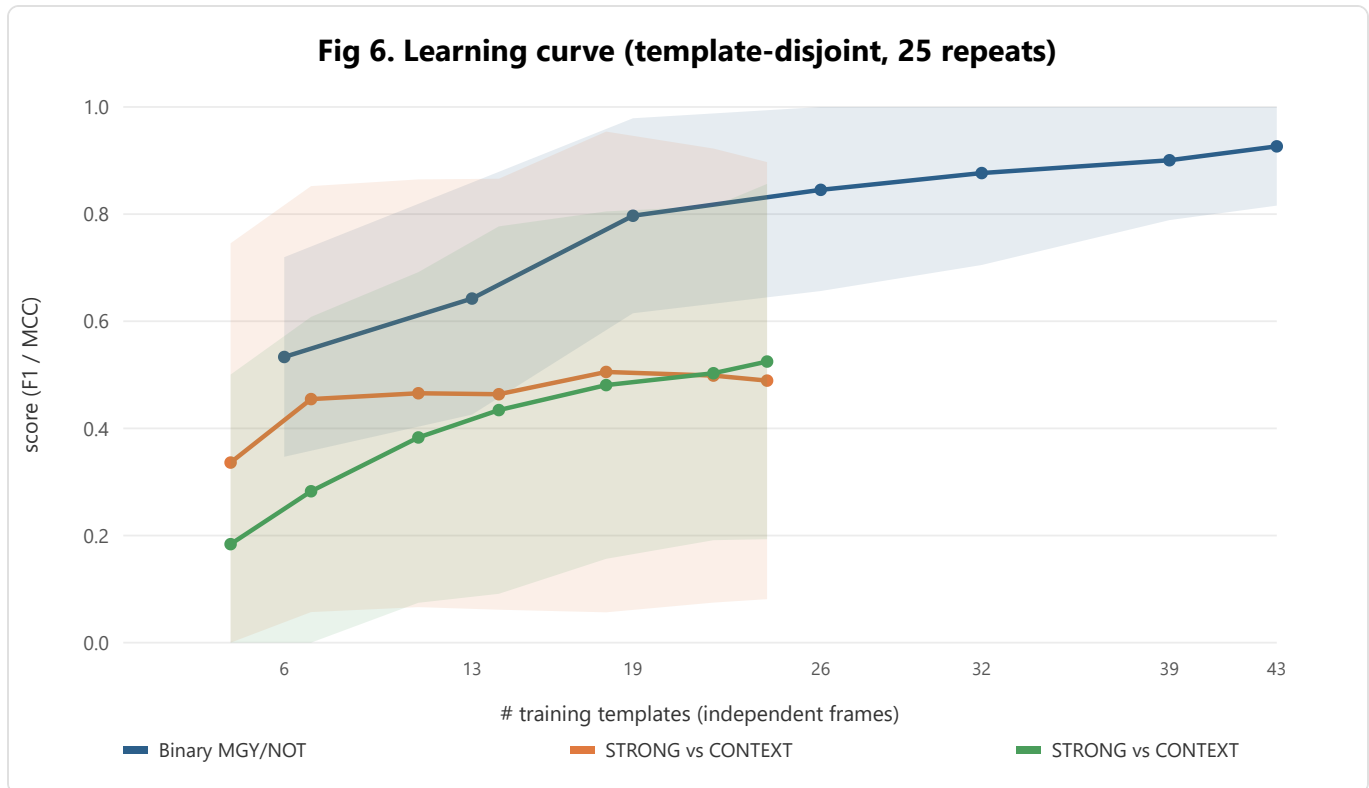


Figure 6 (Appendix A). Learning curve on the found corpus vs number of independent training frames: the binary task is serviceable but unsaturated; the explicit/implicit subtype is data-starved — mean still rising, error band non-shrinking.

The binary task is serviceable but unsaturated (variance shrinks $\pm 0.19 \rightarrow \pm 0.11$); the subtype task is **data-starved** — its mean is *still climbing* at full data and its error band *never contracts* (± 0.32 throughout). This is the empirical reason we move from the found corpus to a purpose-built diagnostic (§5): the explicit/implicit signal in the found data is too thin to measure conclusively, whereas the generator supplies balanced, controlled minimal pairs by design.

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Appendix B — Generator source

The full frame grammar is the generator `generate.js` (seed 20240614). Running it regenerates the dataset along with the category, frame, and pair metadata that the construction-disjoint splits and the consistency metric rely on.